

ROJEAN B. PEÑANO

Software Engineer · Python · AI/Machine Learning · Cloud

Tagaytay City, Cavite, Philippines | +63 956 985 7592 | rojeanpenano@gmail.com
linkedin.com/in/rojeanpenano | github.com/rojeanpenano | Portfolio: rojeanpenano.vercel.app

SUMMARY

Computer Science graduate specializing in Intelligent Systems, focused on Python backend development and applied AI/Machine Learning. Built an award-winning AI farming platform through undergraduate research and developed a serverless service on Microsoft Azure during internship, establishing a strong foundation in scalable system architecture, machine learning integration, and clean software engineering practices. Brings analytical thinking, attention to detail, and a commitment to developing reliable, maintainable, and impactful software solutions.

TECHNICAL SKILLS

Languages: Proficient in Python, JavaScript, HTML, CSS; familiar with Java

Machine Learning & AI: TensorFlow, scikit-learn, Convolutional Neural Networks, image classification, regression analysis, AI model integration into web applications, prompt engineering

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Google Colab

Backend Development: Flask, Express.js, REST API design, JWT authentication, Pydantic and middleware input validation, modular and clean architecture, event-driven programming

Cloud & Databases: Microsoft Azure (Functions, Cosmos DB, Blob Storage), Firebase (Authentication, Cloud Firestore), MongoDB (Mongoose), schema design, multi-source data integration

Tools & Practices: Git / GitHub version control, Postman (API testing), security fundamentals

EXPERIENCE

Backend Developer Intern — Hactiv Colab Inc.

July 2025 – August 2025

- Developed a cloud-based inventory management backend in Python (Flask), deployed serverless on Azure Functions (github.com/rojeanpenano/python-flask-inventory-api).
- Designed modular CRUD REST APIs across five domain blueprints with Pydantic request validation.
- Integrated Azure Cosmos DB and Azure Blob Storage, implementing data storage and retrieval strategies across multiple data sources.
- Implemented transaction integrity across Cosmos DB containers using two-pass validation with compensating rollback, preserving inventory consistency on failed writes.

PROJECTS

Rice Up — Farming Platform (Undergraduate Thesis)

May 2025 – May 2026

github.com/rojeanpenano/rice-up-web

- Engineered a mobile-accessible platform giving Filipino rice farmers AI-driven crop diagnosis, price forecasting, and farming decision support.
- Built a VGG16 transfer-learning model with TensorFlow classifying 9 rice leaf diseases across 10,689 images (76% accuracy), and served it through an isolated Flask REST inference API.
- Diagnosed class imbalance behind an inflated 87% pest-model accuracy through precision-recall analysis; rebalanced and augmented the dataset, raising accuracy to 95% in v2.
- Developed a Node.js linear-regression module for rice price forecasting (5.03% MAPE on official DA price data); implemented “AskJuan,” a structured Filipino-language Q&A assistant, and a farming calendar.
- Developed the backend architecture, database design, and cloud implementation (Firebase Authentication, Cloud Firestore), while thesis teammates led field research and dataset curation.
- Recognition: 1st Place, Technology Exhibit (CONTRENDS 2026) · 2nd Place, Best Thesis · Best Paper Presentation Finalist, ICAI 2026**

XPENDITURE — Expense Management Website (Software Engineering Course Project)

Sep 2024 – Nov 2024

github.com/rojeanpenano/xbudget

- Worked as the backend developer, building REST APIs with Node.js, Express.js, and MongoDB (Mongoose) covering budgets, expense tracking, group budgeting, and transactions.
- Implemented JWT-based user authentication and authorization with middleware-based input validation and error handling, verifying endpoints through API testing with Postman.

EDUCATION

Bachelor of Science in Computer Science, Specializing in Intelligent Systems

De La Salle University–Dasmariñas

2022 – 2026

CERTIFICATIONS

Cisco Networking Academy: Linux Essentials · Operating Systems Basics · Introduction to Cybersecurity · Introduction to JavaScript

WPH Digital: Generative AI Awareness & Effective Prompting · Designing for Humans – UX Psychology

Pearson: Versant English Placement Test